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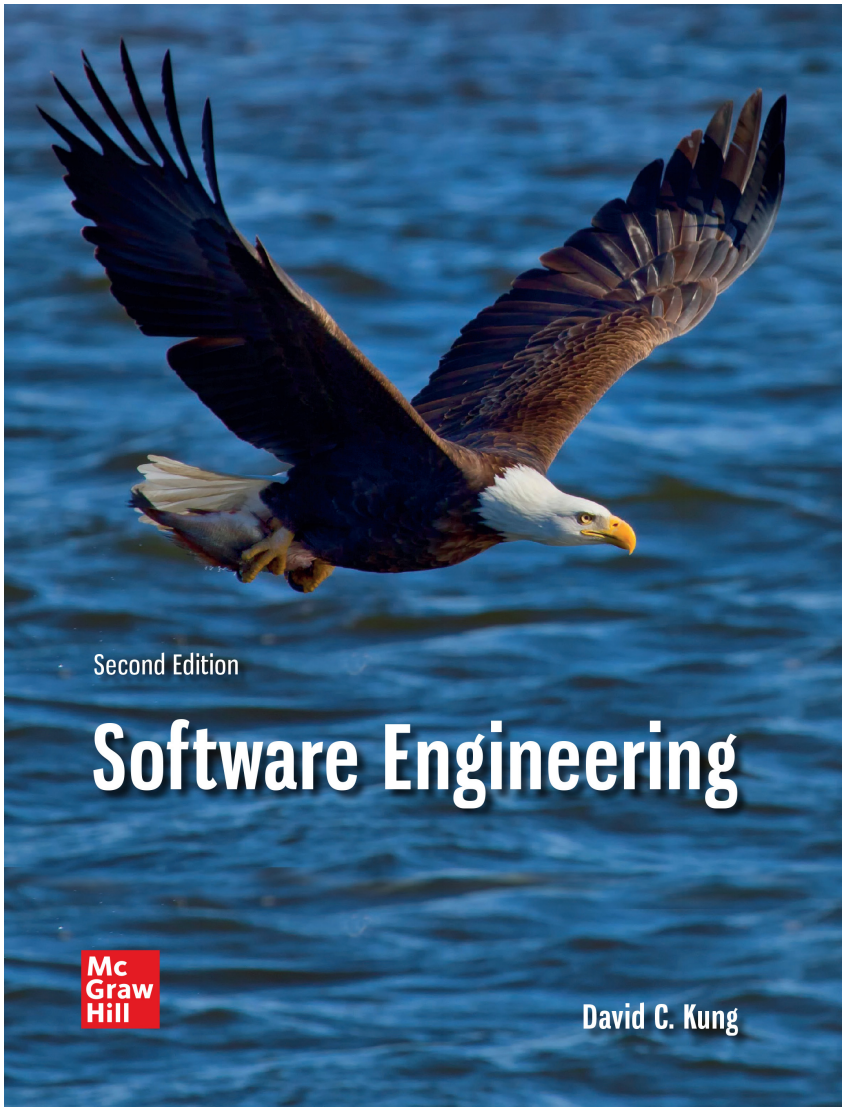
Chapter 02

Process and Methodology

Software Engineering

Second Edition

David C. Kung



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Key Takeaway Points

- A software process defines the phases of activities or what need to be performed to construct a software system.
- A software methodology details the steps or how to perform the activities of a software process. A methodology is an implementation of a process.
- Software development needs a software process and a methodology.

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Software Process

- System development challenges call for an engineering approach for software development. A software process is required.

- **Definition 2.1** A *software process* defines a series of activities performed to construct a software system. Each activity produces some artifacts, which are the input to other phases. Each phase has a set of entrance criteria and a set of exit criteria.

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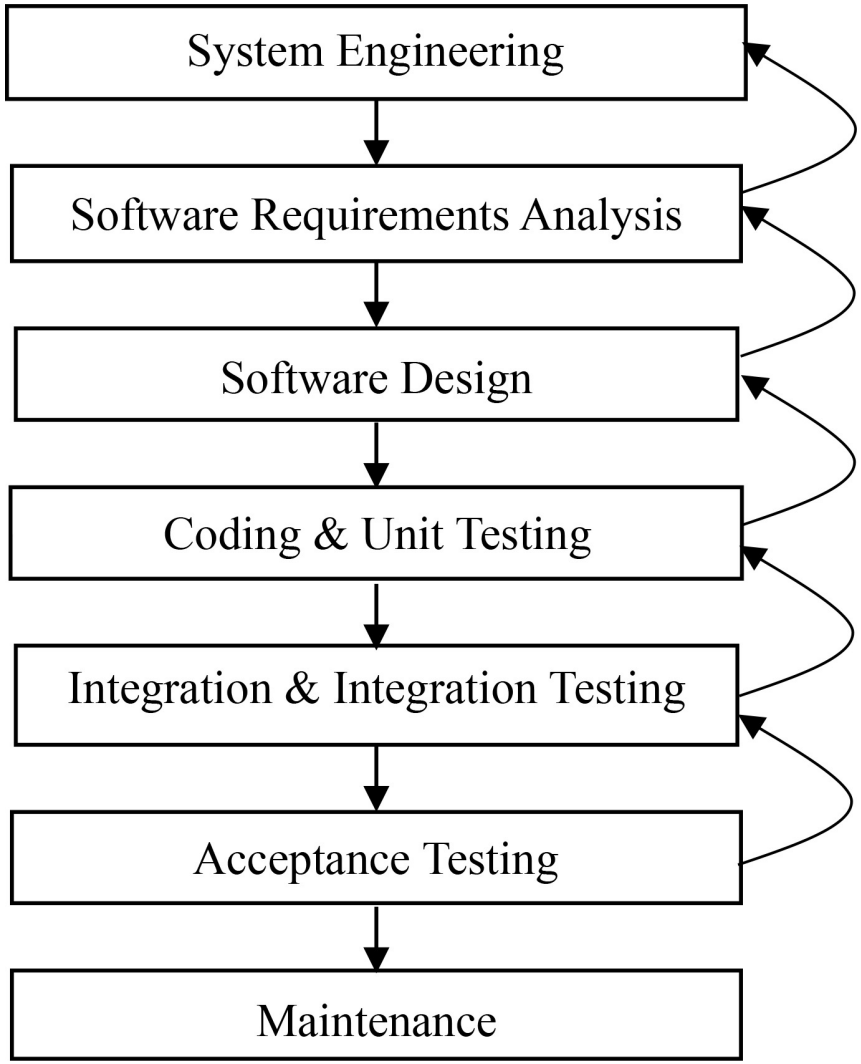
The Waterfall Process



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Merits of the Waterfall Model

The simple, straight sequence of phases of the waterfall simplifies project management.

It supports function-oriented project organization:

- Each project is carried out by a pipeline of functional teams.
- Each functional team is specialized in one function such as requirements analysis, design, implementation, integration and testing, and so forth.

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Problems of the Waterfall Model

- It is inflexible to requirements change.
- The long development duration means the system is outdated when it is delivered.
- Users cannot experiment with the system to provide early feedback.
- The customer has to wait until the entire system is implemented and deployed to reap the benefits.
- The customer may lose the entire investment if the project fails.

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Software Process Models

- Prototyping Process Model
- Evolutionary Process Model
- Spiral Process Model
- Unified Process Model
- Personal Software Process Model
- Team Software Process Model
- Agile Process Models

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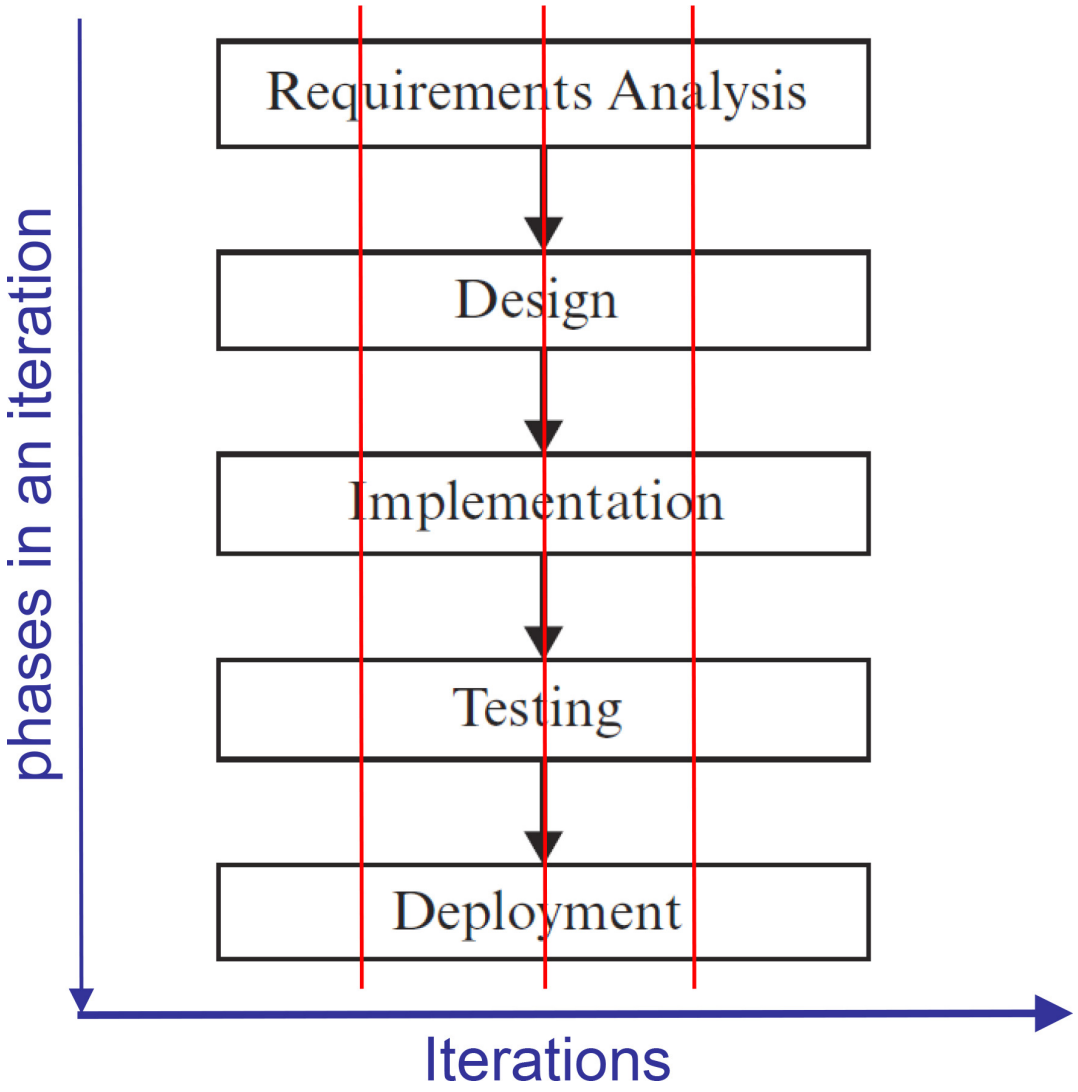
Agile Process Models



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Some Well-Known Agile Methods

- Dynamic Systems Development Method (DSDM)
- Feature Driven Development (FDD)
- Scrum
- Extreme Programming (XP)
- Crystal Clear
- Lean Development

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The Methodology Presented in This Book

- It is designed for beginners as well as seasoned developers.
- It is aimed at educating software architects and systems analysts.
- It can be applied to agile as well as plan-driven projects. It has been applied to sponsored as well as industrial projects.
- Team members should work together from project start to completion.
- Many students continue practicing the methodology after graduation.

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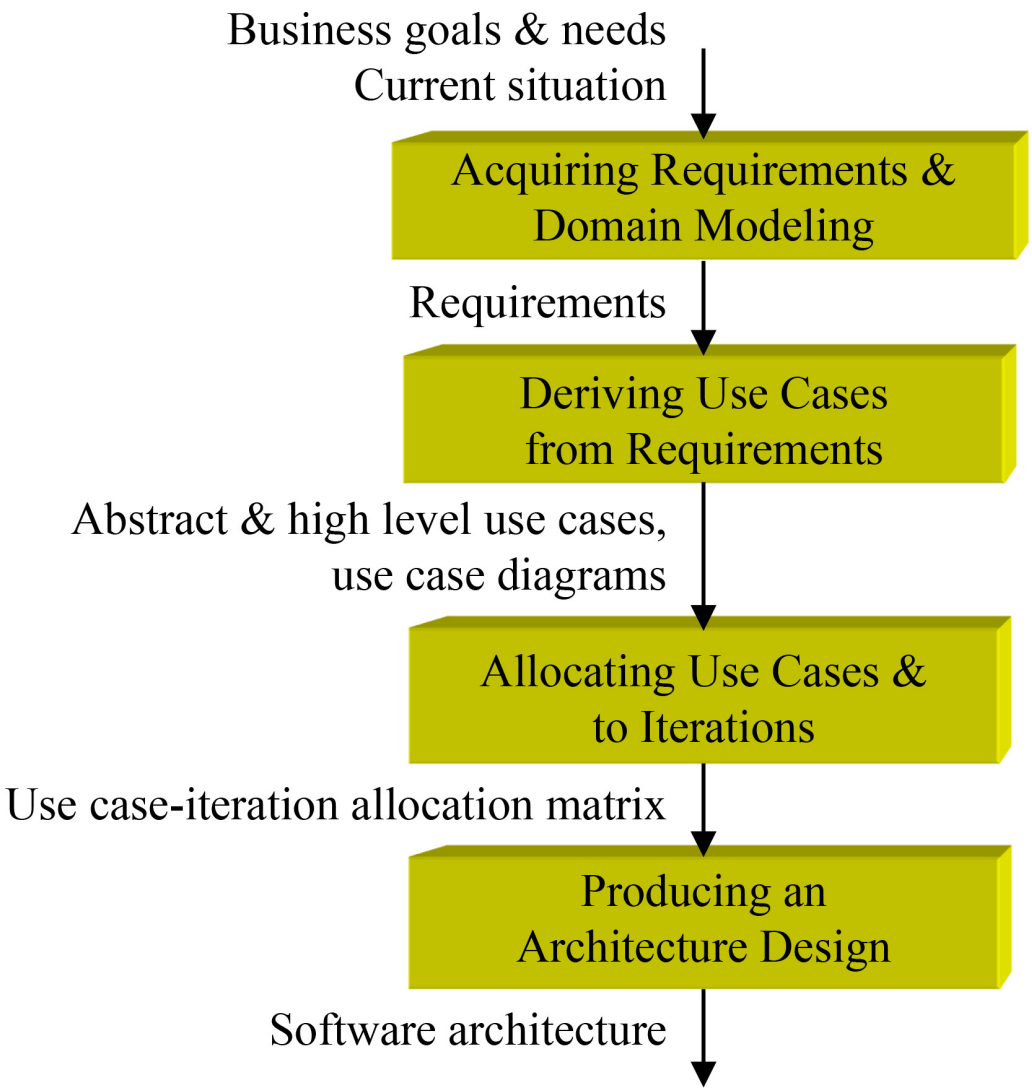
Methodology Overview – Planning Phase



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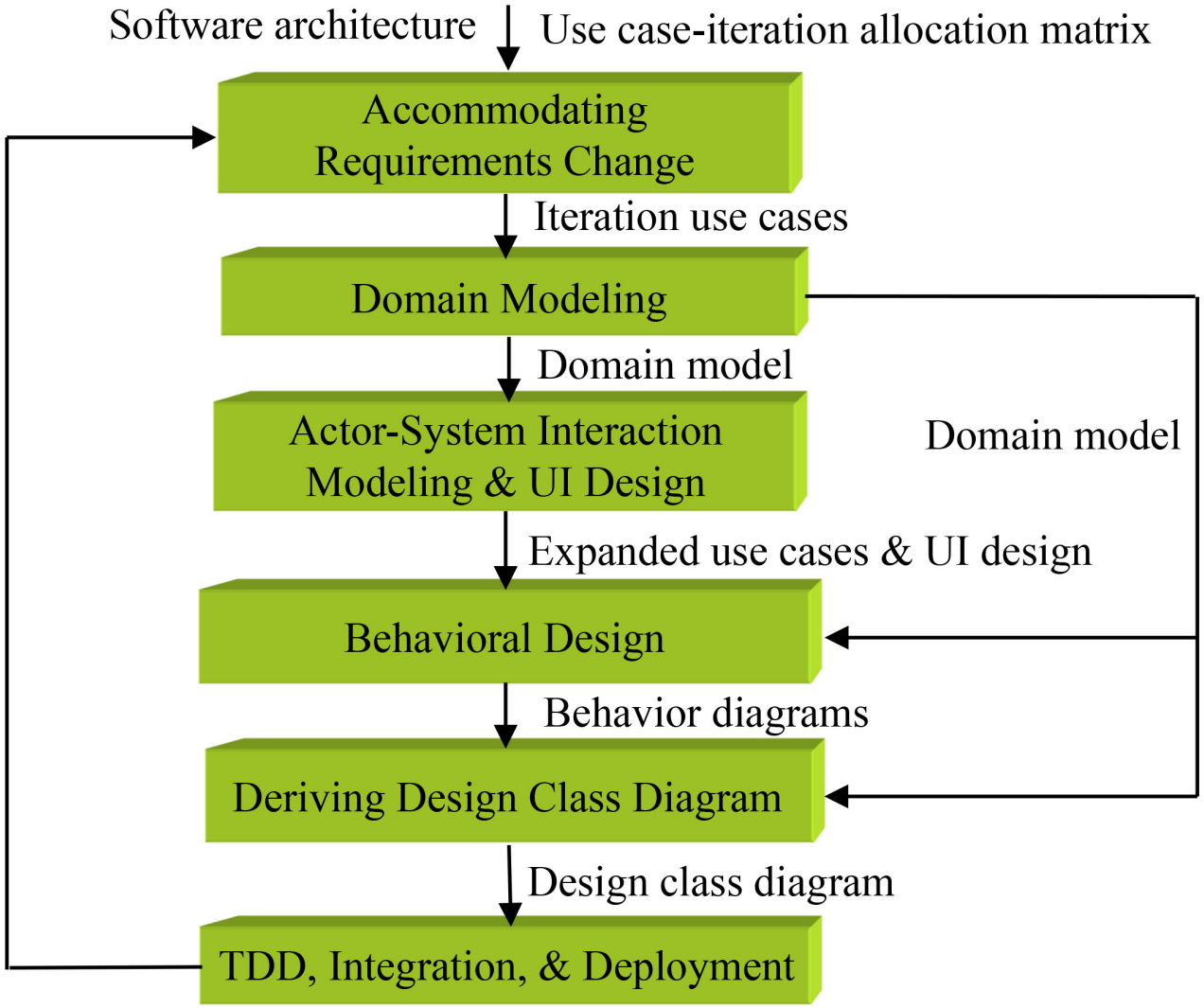
Methodology Overview – Iterative Phase



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